

FORM1 THE PATENTS ACT1970 (39of1970) and THE PATENTSRULES,2003 APPLICATIONFOR GRANT OF PATENT (Seesection7,54and135andsub-rule(1)ofrule20)				(FOROFFICEUSE ONLY)			
				ApplicationNo.			
				Filingdate:			
				AmountofFee paid:			
				CBRNo:			
				Signature:			
1.APPLICANT'SREFERENCE/ IDENTIFICATIONNO.(ASALLOTTEDBY OFFICE)							
2.TYPEOFAPPLICATION[Pleasetick(✓)attheappropriatecategory]							
Ordinary(✓)		Convention()		PCT-NP()		PPH()	
Divisional()	Patentof Addition()	Divisional()	Patentof Addition()	Divisional()	Patentof Addition()		
3A.APPLICANT(S)							
Namein Full	Gender (optional,individuals)	Nationality	Countryof Residence	Age (optional, Fornatural Persons)	AddressoftheApplicant		
SRI VENKATESWARAA COLLEGE OF TECHNOLOGY		INDIAN	INDIA		HouseNo.		
					Street	BHB NAGAR, VADAKAL VILLAGE, ORAGADAM INDUSTRIAL CORRIDOR	
					City	SRIPERUMBUDUR	
					State	TAMIL NADU	
					Country	INDIA	
					Pincode	602105	
					Email(OTP verification mandatory -willbe redacted)		
					Contact number (OTP verification mandatory -willbe redacted)		

3B.CATEGORYOFAPPLICANT[Pleasetick(✓)attheappropriatecategory]				
NaturalPerson()	OtherthanNaturalPerson ()			Educationalinstitution(✓)
	SmallEntity()	Startup()	Others()	
4.INVENTOR(S)[Pleasetick(✓)attheappropriatecategory]				
Arealltheinventor(s)sameasthe applicant(s)namedabove?		Yes ()	No(✓)	

If“No”,furnishthedetailsofthe inventor(s)						
Namein Full	Gender (optional, Fornatural persons)	Nationality	Age(optional, fornatural persons)	Countryof Residence	Addressofthe Inventor	
1. MOHAN ASUNDA RAM R	MALE	INDIAN	39 YEARS	INDIA	HouseNo.	25A
					Street	REDDY STREET
					City	SRIPERUMBUDUR
					State	TAMILNADU
					Country	INDIA
					Pincode	602105
2.PREM ARVINTH G	MALE	INDIAN	21 YEARS	INDIA	HouseNo.	160B/1
					Street	ALAGAR NAGAR
					City	MADURAI
					State	TAMIL NADU
					Country	INDIA
					Pincode	625009
3.CHANDRU S	MALE	INDIAN	21 YEARS	INDIA	HouseNo.	38
					Street	METTU STREET
					City	TIRUVANNAMALAI
					State	TAMIL NADU
					Country	INDIA
					Pincode	604601
5.TITLEOFTHEINVENTION						
Automated Smart Street Light Fault Detection and Response System						
6.AUTHORISEDREGISTEREDPATENT AGENT(S)				IN/PANo.		
				Name		
				MobileNo.(OTP verification mandatory-willbe redacted)		
7.ADDRESSFORSERVICEOFAPPLICANTIN INDIA				SRI VENKATESWARAA COLLEGE OF	BHB NAGAR, VADAKAL VILLAGE, ORAGADAM INDUSTRIAL CORRIDOR, SRIPERUMBUDUR,	

		TECHNOLOGY			
		PostalAddress			
		TelephoneNo.			
		MobileNo.(OTP verificationmandatory-willberedacted)			
		FaxNo.			
		E-mail ID (OTP verificationmandatory-willberedacted)			
8.INCASEOFAPPLICATIONCLAIMINGPRIORITYOFAPPLICATIONFILEDINCONVENTION COUNTRY,PARTICULARSOFCONVENTIONAPPLICATION					
Country	Application Number	Filingdate	Nameofthe applicant	Titleofthe invention	IPC(asclassifiedin theconvention country)
9.INCASEOFPCTNATIONALPHASEAPPLICATION,PARTICULARSOFINTERNATIONAL APPLICATIONFILEDUNDERPATENTCO-OPERATIONTREATY(PCT)					
Internationalapplicationnumber			Internationalfilingdate		
10.INCASEOFDIVISIONALAPPLICATIONFILEDUNDERSECTION16,PARTICULARS OF ORIGINAL(FIRST)APPLICATION					
Original(first)applicationNo.			Dateoffilingoforiginal(first)application		
11.INCASEOFPATENTOFADDITIONFILEDUNDERSECTION54,PARTICULARSOFOF MAIN APPLICATION OR PATENT					
Mainapplication/patentNo.			Dateoffilingofmainapplication		
12. DECLARATIONS					

(i) Declarationbytheinventor(s)

(Incasetheapplicantisanassignee:theinventor(s))may signhereinbelowortheapplicantmayuploadtheassignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

I/We,theabovenamedinventor(s)is/arethetrue&firstinventor(s)forthisInventionanddeclarethattheapplicant(s) herein is/are my/our assignee or legal representative.

(a) Date : 29-04-2026

(b) Signature(s)

(c) Name(s) : Payal Soni Gupta

(a) Date : 29-04-2026

(b) Signature(s)

(c) Name(s) : Prem Arvinth G

(a) Date : 29-04-2026

(b) Signature(s)

(d) Name(s) : Chandru S

(ii) Declaration by the applicant(s) in the convention country

(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date

(b) Signature(s)

(c) Name(s) of the signatory

(iii)Declarationbytheapplicant(s)

I/We the applicant(s) hereby declare(s) that:-

- ☐ I am/We are in possession of the above-mentioned invention.
- ☐ The provisional/complete specification relating to the invention is filed with this application.
- ☐ The invention as disclosed in the specification uses the biological material from India, and the necessary permission from the competent authority shall be submitted by me/us before the grant of the patent to me/us.
- ☐ There is no lawful ground of objection(s) to the grant of the patent to me/us. I
- ☐ am/We are the true & first inventor(s).
- ☐ I am/We are the assignee or legal representative of the true & first inventor(s).
- ☐ The application or each of the applications, particulars of which are given in Paragraph-8, was the first application in convention country/countries in respect of my/our invention(s).
- ☐ I/We claim the priority from the above-mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from whom I/We derive the title.
- ☐ My/our application in India is based on an international application under the Patent Cooperation Treaty (PCT) as mentioned in Paragraph-9.
- ☐ The application is divided out of my/our application, particulars of which are given in Paragraph-10, and I/we pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under Section 16 of the Act.
- ☐ The said invention is an improvement in or modification of the invention, particulars of which are given in Paragraph-11.

13.FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
complete	No. of pages- 1		
No. of Claim(s)-03	No. of claims and No. of Pages		
Abstract	No. of pages		
No. of Drawing(s)- 1	No. of drawings and No. of Pages		

#Incaseofacompletespecification,iftheapplicantdesirestoadoptthedrawingsfiledwithhisprovisionalspecification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of Suchpagesfiledwiththeprovisionalspecificationarerequiredtobementionedhere.

- (b) Completespecification(inconformationwiththeinternationalapplication)/asamendedbeforethe International Preliminary Examination Authority (IPEA), as applicable (2 copies).
(c) Sequencelistinginelectronicform.
(d) Drawings(inconformationwiththeinternationalapplication)/asamendedbeforetheInternational Preliminary Examination Authority (IPEA), as applicable (2 copies).
(e) Prioritydocument(s)orarequesttoretrievetheprioritydocument(s)fromDAS(DigitalAccessService)ifthe applicant had already requested the office of first filing to make the priority document(s) available to DAS.
(f) Translationofprioritydocument/Specification/InternationalSearchReport/InternationalPreliminary Report on Patent ability.
(g) StatementandUndertakingonForm 3.
(h) DeclarationofInventorshiponForm5.
(i) Power of Authority.
(j).....

Totalfee₹.....inCash/Banker'sCheque/BankDraftbearingNo..... Date.....on.....
Bank.

I/Weherebydeclarethattothebestofmy/ourknowledge,informationandbeliefthefactandmattersslatedherein are correct and I/We request that a patent maybe granted to me/us for the said invention.

Datedthis.....dayof.....20..... Signature:
Name:

To,
TheControllerof Patents
ThePatentOffice,at.....

Note:-

- Repeatboxesincaseofmorethanoneentry.
- Tobesignedbytheapplicant(s)orbyanauthorizedregisteredpatentagent,otherwisewherementioned.
- Tick(✓)/cross(x)whicheverisapplicable/notapplicableinthedeclarationinParagraph-12
- Nameoftheinventorandapplicantsshouldbegiveninfull,familynamethebeginning.
- Strikeouttheportionwhichis/arenotapplicable.
- Forfee:SeeFirstSchedule.

FORM2

**THE PATENT ACT 1970 (39
OF 1970)**

&

THE PATENTS RULES, 2003

PROVISIONAL/COMPLETE SPECIFICATION

(SEE SECTION 10 AND RULE 13)

1. TITLE OF THE INVENTION:

Smart home energy management system with predictive analytics and adaptive load optimization using ANN

2. APPLICANT(S)

Name	Nationality	Address
SRI VENKATESWARAA COLLEGE OF TECHNOLOGY		BHB NAGAR, VADAKAL VILLAGE, ORAGADAM INDUSTRIAL CORRIDOR, SRIPERUMBUDUR - 602105

3. REAMBLE TO THE DESCRIPTION

PROVISIONAL	COMPLETE
The following specification describes the	the following specification invention. Particularly describes the invention and the manner in which it is to be performed.

4 DESCRIPTIONS

The present invention relates to an intelligent street lighting system designed for efficient energy management and fault detection in public lighting infrastructure. The system utilizes real-time monitoring, current sensing, and automated control using a microcontroller (ESP32) to optimize energy consumption, detect faulty street lights, and improve maintenance efficiency. It integrates sensors such as LDR (Light Dependent Resistor) and current sensors to automate lighting based on environmental conditions and electrical parameters.

1 Objective of The Invention

1. To develop an automated street lighting system that controls ON/OFF operation based on ambient light conditions using LDR, reducing unnecessary energy consumption.
2. To monitor current flow in street lights using current sensors and detect faulty or non-working lights in real time.
3. To reduce power wastage by ensuring lights operate only when required and identifying abnormal current variations.
4. To provide real-time data monitoring and fault alerts using ESP32 communication, enabling efficient maintenance and reduced manual inspection.

2 System Architecture:

The smart street light monitoring system consists of the following components:

Central Controller (ESP32): A microcontroller that collects data from sensors (LDR and current sensor), processes it, and transmits information to a monitoring system.

Smart Devices: These include smart thermostats, lighting systems, HVAC units, kitchen appliances, and other energy-consuming devices that are connected to the controller via Wi-Fi, Zigbee, Bluetooth, or other wireless communication protocols.

LDR Sensor: Detects ambient light intensity and controls automatic switching of street lights (ON at night, OFF during daytime).

Renewable Energy Sources: Solar panels, wind turbines, or other renewable generation units connected to the system, which provide energy to the home, reducing reliance on the grid.

Current Sensor (ACS712 or similar): Measures current flowing through street lights to determine operational status and detect faults.

Power Supply Unit: Includes a step-down transformer and rectifier circuit to convert AC supply (230V) to required DC voltage (e.g., 12V and 3.3V).

Communication Module: Built-in Wi-Fi in ESP32 used to send data to server/cloud for monitoring.

3 Key features of the system include:

Automatic Light Control – Street lights turn ON/OFF based on ambient light using LDR.

Fault Detection – Identifies non-working lights using current variations.

Energy Efficiency – Reduces unnecessary power consumption.

Real-Time Monitoring – Sends data only when current fluctuation is detected and periodically at night.

Scalability – Can be extended to multiple street light networks.

Low Maintenance Cost – Reduces manual inspection.

4 Operation

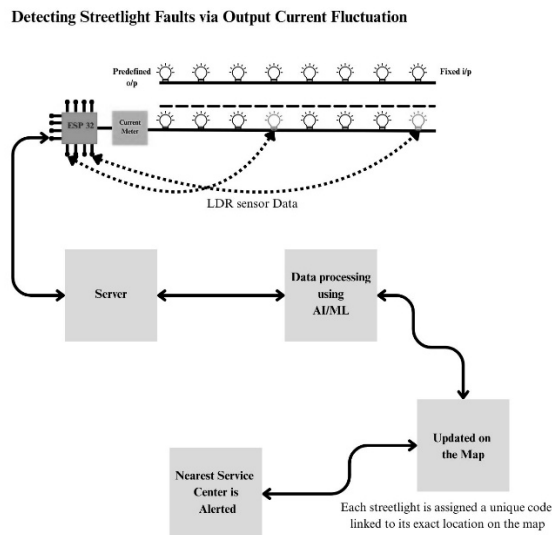
The system operates by continuously gathering data from connected sensors including the LDR and current sensor, along with real-time electrical parameters from the street lighting network. Based on the detected ambient light intensity, the system determines the time of day, wherein during low light conditions the controller activates the street lights and during high light conditions the lights remain in OFF state. The controller further monitors the current consumption of each street light and compares it with predefined threshold values to evaluate the operating condition. This monitoring mechanism enables the system to optimize street light operation by ensuring lights function only when required, thereby minimizing energy consumption. Additionally, the system prioritizes efficient energy usage by avoiding unnecessary operation during daytime and maintaining proper functioning during nighttime. The system can also communicate with remote monitoring platforms through IoT-based communication, allowing real-time fault detection, status updates, and efficient management of the street lighting infrastructure during varying operational conditions.

5 Smart Street Light Monitoring System

The system provides an intelligent solution for street lighting by combining sensing, control, and communication technologies. It uses real-time current monitoring and light detection to ensure efficient operation. The ESP32 controller processes sensor data and identifies faulty lights based on predefined current thresholds.

The system minimizes energy consumption by automating lighting and reduces maintenance efforts by pinpointing exact faulty lights. It also enables remote monitoring, making it suitable for smart city applications.

6 Block Diagram



7 Summary of the Invention

The present invention relates to a smart home energy management system designed to improve energy efficiency, reduce utility costs, and enhance user comfort through the use of predictive analytics and adaptive load optimization. The system integrates smart appliances, renewable energy sources, and grid-based energy management to optimize energy consumption. It addresses the limitations of traditional energy management systems by dynamically adjusting to changing consumption patterns, external energy prices, and renewable energy generation, ensuring sustainability and cost savings.

The system leverages machine learning algorithms to predict energy demand based on historical data, weather forecasts, and grid pricing signals. It optimizes the operation of appliances, heating, cooling, and lighting by shifting energy-intensive tasks to off-peak hours, prioritizing renewable energy when available, and participating in demand-response programs to reduce peak consumption. The integration of renewable energy sources such as solar panels and wind turbines helps to reduce reliance on grid power and enhance energy sustainability.

Additionally, the system provides users with real-time insights into their energy consumption, offering actionable recommendations for reducing waste and saving on utility costs. Through a mobile or web application, users can monitor and control appliances, making the system highly customizable to meet the specific needs of each household. This invention enables efficient energy use, enhances comfort, and contributes to the reduction of environmental impact.

8 ClaimSection(CompleteSpecification)

1. A smart street light monitoring system comprising a central controller that communicates with various components within the street lighting network, including LDR sensors, current sensors, street lights, and communication modules, to collect real-time data and monitor lighting conditions based on ambient light intensity and electrical parameters.
2. The system of claim 1, wherein the central controller automatically controls the operation of street lights based on ambient light intensity detected through the LDR sensor, ensuring that the lights are switched ON during low light conditions and OFF during high light conditions, thereby minimizing energy consumption.
3. The system of claim 1, wherein current sensing is employed to continuously monitor the current flowing through each street light, enabling detection of faulty, damaged, or non-operational lights by identifying deviations from predefined current threshold values.
4. The system of claim 1, wherein the system detects faults in real time by analyzing current fluctuations and abnormal operating conditions, thereby reducing the need for manual inspection and improving maintenance efficiency.
5. The system of claim 1, wherein the controller transmits data only when abnormal current variation or fault condition is detected, thereby reducing communication overhead and optimizing system performance.
6. The system of claim 1, further comprising a communication module that enables real-time monitoring and reporting of street light status through IoT-based platforms, allowing remote access and control of the lighting infrastructure.
7. The system of claim 1, wherein the controller is configured to send periodic status reports of all street lights at predefined intervals, enabling efficient monitoring, analysis, and maintenance of the street lighting system.

ABSTRACT

Street lighting is a critical component of urban infrastructure, ensuring road safety, reducing accidents, and enhancing public security during night hours. However, traditional streetlight maintenance methods rely heavily on manual inspection and public complaints, which often result in delayed fault detection, increased maintenance costs, and unnecessary energy wastage. Although some modern systems implement IoT-based monitoring, they typically adopt pole-wise architectures that require individual sensors and communication modules for each streetlight. This approach increases hardware cost, installation complexity, and communication overhead. This project proposes a centralized row-wise monitoring architecture using an ESP32 microcontroller for efficient streetlight fault detection. Instead of equipping every pole with separate monitoring units, a group of streetlights is supervised collectively through shared sensing modules. The system continuously measures key electrical parameters such as voltage, current, and brightness level to identify faults including lamp failure, overload, short circuit, and abnormal operating conditions. When a fault is detected, an instant alert is transmitted through IoT communication to the monitoring authority. The proposed system reduces hardware complexity, minimizes network traffic, and enhances scalability for large-scale deployment. It is suitable for urban, semi-urban, and rural environments and provides reliable real-time fault detection, thereby improving maintenance efficiency, energy management, and public safety.

Keywords— ESP32, Smart Street Light, Fault Detection, IoT, LDR Sensor, Voltage Sensor, Current Sensor, Embedded C, Arduino IDE.

FORM3

THE PATENTS ACT, 1970

(39 of 1970)

and

THE PATENT RULES, 2003

STATEMENT AND UNDERTAKING UNDER SECTION 8

(See sub-rule (2) and (3) of Rule 12)

1. Name of the applicant(s).			I/We Sri Venkateswaraa College of Technology Hereby declare:		
2. Name, address and nationality of the joint applicant. ECR, poonjeri, Mamallapuram, Chennai: 603604 Tamilnadu India			(i) That I/We who have made the application for patent number.....in India, dated....., alone/jointly with..... , (ii) that I/We have not made any application for the same/substantially the same invention outside India Or (iii) that I/We have made for the same/substantially same invention, application(s) for patent in the other countries, the particulars of which are given below:		
Name of the country	Date of application	Application No.	Status of the application	Date of publication	Date of disposal
3. Name and address of the assignee			(i) that the rights in the application(s) filed in India has/have been assigned to (ii) that I/We undertake that up to the date of grant of the patent by the Controller, I/We would keep him informed in writing regarding the details of corresponding applications for patents filed outside India in accordance with the provisions contained in section 8 and rule 12. Dated this.....day of20.....		
4. To be signed by the applicant or his authorized registered patent agent.			Signature.....		

5.Nameofthenaturalpersonwhohas signed.	(.....)
	To TheControllerof Patents, ThePatentOffice, Chennai
Note.-Strikeoutwhicheverisnotapplicable.	

FORM 5
THE PATENTS ACT, 1970 (39
of 1970)

The Patents Rules, 2003

DECLARATION AS TO INVENTORSHIP
[See section 10(6) and rule

1. NAME OF APPLICANT
(S)

Sri Venkateswaraa College of Technology

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of my/our application numbered dated _____ is/are _____

2. INVENTOR(S)

1. NAME: MOHANASUNDARAM
NATIONALITY: INDIAN

ADDRESS: 28A, REDDY STREET, SRIPURUMBUDUR - 602105

Dated this day of 20

Signature: -

Name of the signatory: -

2. NAME: G. PREM ARVINTH
NATIONALITY: INDIAN

ADDRESS: 160B/1, ALAGAR NAGAR, KAMARAJAR SALAI, MADURAI - 625009

Dated this day of 20

Signature: -

Name of the signatory: -

3. NAME : S. CHANDRU
NATIONALITY: INDIAN

ADDRESS: 38, METTU STREET, KARUNGALIKUPPAM VILLAGE, TIRUVANNAMALAI - 604601

Dated this day of 20

Signature: -

Name of the signatory: -

3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS FILED BY THE
APPLICANT (S) IN THE CONVENTION COUNTRY:

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this day of 20

Signature: -

Name of the signatory: -

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form) I/We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

Dated this.....day of.....20.

Signature of the additional inventor(s): Name:-

To, The Controller of Patent The
Patent Offices

FORM9
THE PATENTS ACT, 1970 (39
OF 1970)

&

The Patents Rules, 2003

REQUEST FOR PUBLICATION
[See section 11A(2); rule 24A]

1. Name, Address and nationality of the
applicant(s)

We Sri Venkateswaraa College of Technology, BHB
Nagar, Vadakal Village, Oragadam Industrial Corridor,
Sriperumbudur, Tamil Nadu - 602105

2. To be signed by the applicant or his
authorized registered patent agent

Hereby request for early publication of my/our application
for patent No
..... Dated under
section 11A(2) of the Act

Dated this day of 2025

3. Name of the natural person who has
signed

Signature

.....

To
The Controller of Patents, The Patent Office
Chennai

“FORM28
THEPATENTSACT,1970(39of1970)
AND
THEPATENTSRULES,2003
TOBESUBMITTEDBYASMALLENTITY/STARTUP/EDUCATIONALINSTITUTION
[Seerules2(fa),2(fb),2(ca)and7]

1	Insertname, address and nationality.	I/We Sri Venkateswaraa College of Technology Applicant/patenteeinrespectofthepatentapplicationno..... or patent no..... Hereby declare that I/we am/are a small entity in Accordancewithrule2(fa)orastartupinaccordance withrule2(fb)oraneducationalinstitutioninaccordancewithrule2(ca)andsubmit the following document(s) as proof: Sri Venkateswaraa College of Technology
2	Documentstobesubmitted	
	i.Forclaimingthestatusofasmallentity:	
	A.ForanIndianapplicant:EvidenceofregistrationundertheMicro,SmallandMedium EnterprisesAct,2006(27of2006).	
	B.Incaseofaforeignentity:Anyother document.	
	ii.Forclaimingthestatusofastartup	
	A.ForanIndianapplicant:Anydocumentasevidenceofeligibility,asdefinedinrule2(fb).	
	B.Incaseofaforeignentity:Anyotherdocument.	
	iii.Forclaimingthestatusofaneducationalinstitution	
	A.ForanIndianapplicant:Anydocumentasevidenceof eligibility	
	B.Incaseofaforeigneducationalinstitution:Anyother document.	
3	To be signed by the applicant(s) / patentee(s)/authorizedregisteredpatentagent.	Theinformationprovidedhereiniscorrecttothe best of my/our knowledge and belief. Datedthis.....dayof.....20...
4	Name of the natural person who has signed Designationandofficialseal,ifany,oftheperson who has signed.	(Name)..... (Signature)..... (Designation)..... To The Controller of Patents Thepatentoffice,Chennai

